

# Machine Learning Augmented TCAD Assessment of Corner Radii in Nanosheet FET

<sup>a</sup>Jyoti Patel, <sup>b</sup>Bathula Satwik, <sup>\*b</sup>Navjeet Bagga, <sup>a</sup>Ishani Bais, <sup>a</sup>Chirag Arora, <sup>c</sup>Vivek Kumar, <sup>d</sup>Ankit Dixit, <sup>d</sup>Naveen Kumar, <sup>d</sup>Vihar Georgiev and <sup>a</sup>S. Dasgupta

<sup>a</sup> Indian Institute of Technology Roorkee, India, <sup>b</sup> Indian Institute of Technology Bhubaneswar, Odisha, India, <sup>c</sup> National Institute of Technology, Uttarakhand, India, University of Glasgow, United Kingdom (<sup>\*</sup>navjeet@iitbbs.ac.in)

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## ABSTRACT

In this paper, we proposed a machine learning approach to assist the TCAD results in realizing a local cost and time-effective simulator for analyzing the performance metric of the vertically stacked Nanosheet FET (NSFET). The corners are responsible for field crowding inside the sheets, which significantly affects the parasitic capacitance and thereby reduces the  $I_{ON}/C_{gg}$  ratio. Thus, a detailed insight into corner radii optimization is worth needed. We used Sentaurus TCAD to obtain the results and further realized a local simulator using an XGBoost model to analyze process variations and the role of uneven radii corners in NSFET. In addition, a data augmentation strategy is proposed that leverages the powers of stacked autoencoders (SAE) and InfoGANs to enhance data generalization, improving model robustness and predictive reliability.

## 1. Introduction

The semiconductor industry's growth depends on how the devices are efficiently designed for optimal and reliable performance. Thus, the amalgamation of modern computational learning tactics, such as deep learning, reinforcement learning, etc. with devices may offer an implicit way to perform a cost and time-effective solution [1]. Recently, NSFETs have emerged as potential successors to FinFETs owing to their excellent gate controllability and increased effective width per footprint area [2]-[3]. One prime challenge in NSFET fabrication is the reliability of each sheet/stack. Even slight variations in sheet geometry, particularly the corners, significantly impacted the device characteristics. Sharp corners can lead to localized electric field crowding, thereby increasing leakage current and reducing reliability [4]. Thus, a faster and more efficient approach is needed for the variability analysis. One possible way is by employing modern simulators, where numerous models are present to govern the corner rounding effect; however, the accuracy and computational speed are debatable for actual/realistic device design flow. Further, predicting the corner rounding effect and fault analysis is difficult in commercial simulators. Thus, the data augmentation technique can be employed to improve the simulation efficiency for non-planar devices (i.e., NSFET) by incorporating synthetic data. The utilization of SAE and InfoGANs enhances the data quality, thus, facilitating accurate predictions [5].

## 2. TCAD Setup and Machine Learning Model:

Sentaurus TCAD [6] is used to simulate a single stack three-sheet NSFET as a baseline reference (Fig.1a-b). The simulation setup is calibrated against the experimental data [7] (Fig.2c). The TCAD setup includes a drift-diffusion

and quantum potential model for carrier transport. The thin layer and high field saturation mobility models for governing the mobility degradation in thin sheets. The Auger and SRH model for carrier recombination; K. P model for energy band shift, etc. are included in the TCAD setup. The machine learning (ML) model is designed using the scalable XGBoost algorithm, employed for supervised learning. We used root mean square error (RMSE) as a loss function to get the optimum accuracy. For data augmentation, the stacked autoencoder with InfoGANs is employed to generate the data which follows the distribution of the original data by maximizing the mutual information.

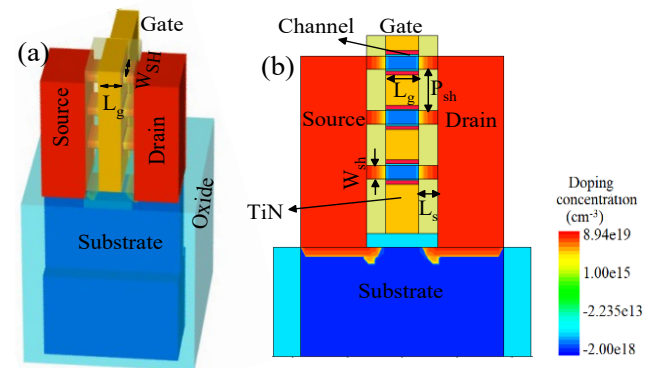


Fig. 1. (a) a three-dimensional (3D) schematic of baseline NSFET; (b) a cross-sectional view, showing all the dimensions of the device.

Table 1: Device Parameter

| Parameter                 | Value                     |
|---------------------------|---------------------------|
| Gate length ( $L_g$ )     | 12 nm                     |
| Eff. oxide thick.         | 0.9 nm                    |
| Extension ( $L_{SP}$ )    | 5 nm                      |
| Sheet Thickness           | 5 nm                      |
| Sheet Spacing             | 11 nm                     |
| Source/drain (S/D) doping | $10^{20} \text{ cm}^{-3}$ |
| Channel doping            | $10^{16} \text{ cm}^{-3}$ |

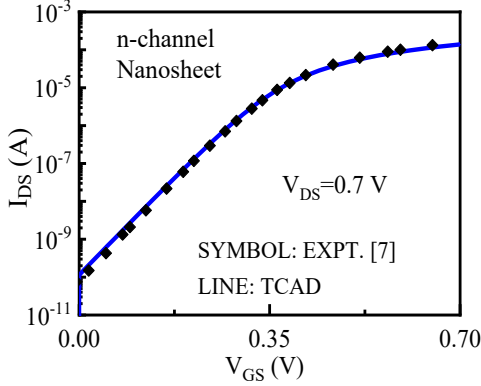


Fig. 2. TCAD model calibration against the experimental data [7]. Table-1 comprises the parameters used in the simulation.

### 3. Results and Discussion:

In NSFET, the electrostatic favorable regions are the corners, and the rounding uniformly distributes the electrons along the side walls. Corner radii (Fig.3a) significantly impact carrier mobility, as the carriers are confined in narrow sheets. Fig 3 (b-d) shows the electrical characteristics of the NSFET for the different corner radii ( $R$ ).

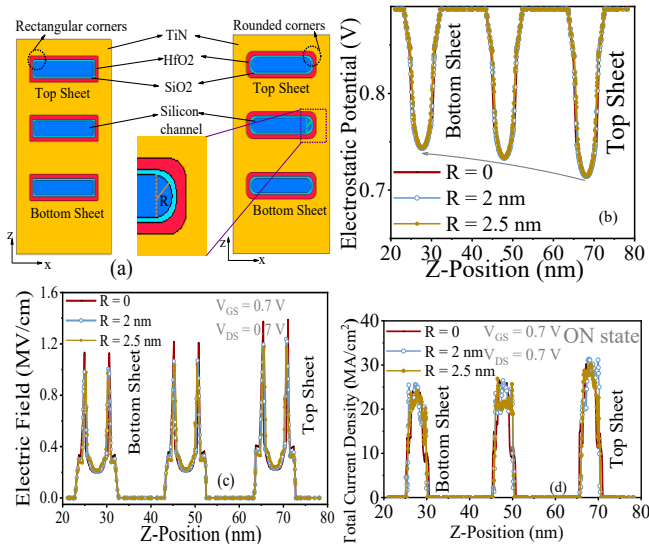


Fig. 3. (a) Cross-sectional view (along the x-z axis) of NSFET for a rectangular and rounded corner. The plots are drawn for different sheets, and the cutline is taken at the corner surfaces: (b) electrostatic channel potential; (c) electric field; (d) ON state current density.

The electric field at the corners sharply increases in a rectangular-shaped sheet; however, it gets evenly distributed for rounded corners, which increases the field flux and, in turn, reduces the OFF current. The maximum electric field is observed at the sharp edges, whereas, for rounded corners of radius ( $R$ ), the field reduces (Fig.3c). Thus, electron mobility and drift velocity increases at the corner.

Moreover, the potential difference from the top to bottom sheet shows the weak gate control in the bottom sheets (Fig.3b). Thus, the total current density shows a slight improvement in rounded corners (Fig.3d). However, leakage increases while moving from the top to the bottom sheet due to reduced gate control for the bottom sheet.

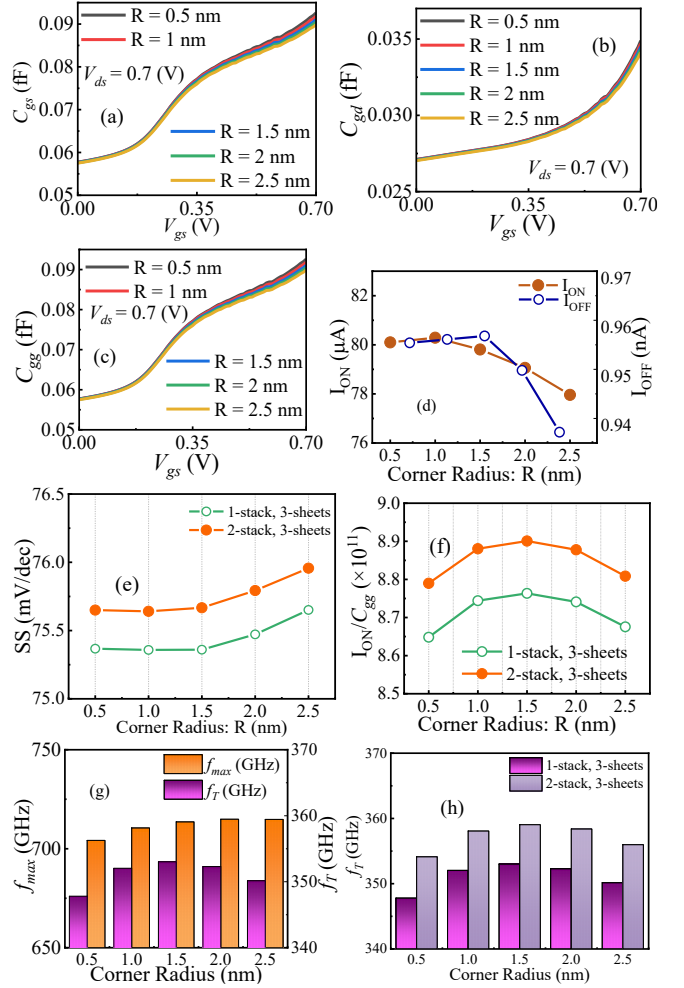
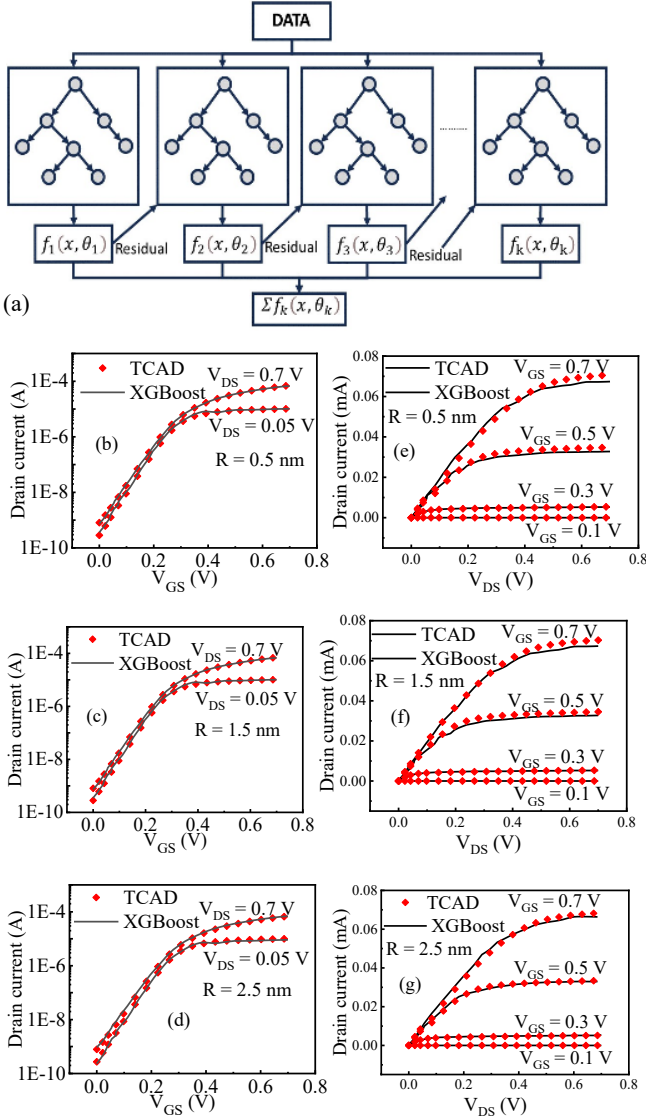


Fig. 4. (a, b, c) The capacitances for various  $R$  values,  $C_{gs}$ ,  $C_{gd}$ , and  $C_{gg}$  with varying gate voltage. (d)  $I_{ON}$  and  $I_{OFF}$  variation with corner radius ( $R$ ) (e) subthreshold variation with corner radii; (f) Iso- $I_{ON}/C_{gg}$  ratio with varying  $R$  for single and double stack NSFET, (g)  $f_T$  and  $f_{max}$  for different  $R$ ; (h) cut-off frequency for single and double stack.

Fig. 4 shows the DC and RF behavior of NSFET to realize the optimized  $R$ . OFF-current ( $I_{OFF}$ ) reduces with increasing corner radii at the cost of reduced ON-current ( $I_{ON}$ ). The maximum  $I_{ON}/I_{OFF}$  ratio is obtained at  $R=1 \text{ nm}$  (Fig.4d). However, gate capacitance, i.e., gate-to-source capacitance ( $C_{gs}$ ), gate-to-drain ( $C_{gd}$ ), and total gate capacitance ( $C_{gg}$ ), decreases with the increasing  $R$  (Fig.4a-c). Thus, a tradeoff between the  $I_{ON}$  and  $C_{gg}$  offers an optimum design guideline for iso- $I_{ON}/C_{gg}$ . The maximum  $I_{ON}/C_{gg}$  ratio (Fig.4f) is achieved at  $R=1.5 \text{ nm}$ , in turn, offering the maximum cut-off frequency ( $f_T$ ) of 353 GHz for the same radii (Fig.4g). However, the maximum frequency of oscillation ( $f_{max}$ ) shows a slight increase with increased rounding. In TCAD simulation, symmetric corner variations have been considered, resulting in even charge and electric field distributions throughout the sheet surface. Further, we

simulated the two stacked three-sheet NSFET device for performance comparison in terms of subthreshold slope (Fig.4e),  $f_T$  (Fig.4h), and  $I_{ON}/C_{gg}$  (Fig.4f).



**Fig.5. (a) XGBoost used as a regressor. The boosting rounds have been set to 100, the learning rate to 0.1, and the max depth to 3; (b-g) Model calibration with TCAD data.**

To acquire a time-efficient solution of such second-order effects in non-planar geometrical structures, we proposed an ML approach using the XGBoost algorithm for optimal analysis and accurate prediction. The data preprocessing part plays a crucial role in acquiring the optimal performance of the model. The Box-Cox transformation [T( $I_{DS}$ )] has been implemented on the TCAD data ( $I_{DS}$ - $V_{GS}$ ) to stabilize the data variance and facilitate its approximation to a normal distribution.

$$T(Y) = \begin{cases} \frac{Y^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0, \\ \log(Y) & \text{if } \lambda = 0, \end{cases} \quad (1)$$

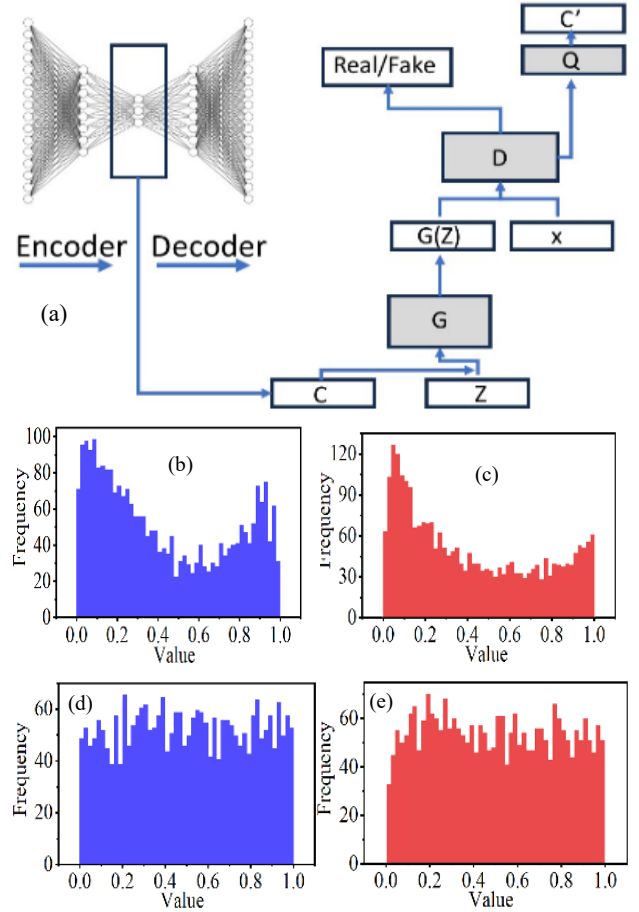
The data is subjected to a modification using  $\lambda = 0.1695$ . Using the XGBoost regressor, the regression has been done (Fig.5a), which stemmed from multiple merits like parallel processing capabilities and its capability to capture nonlinear relationships in the data. Its built-in regularization term aids in reducing overfitting. The boosting rounds have been set to 100, learning rate has been set to 0.1. The predictions have been Gaussian filtered with a  $\sigma = 0.5$  and then the lowess filter has been applied and smoothed with a fraction of 0.1. The proposed XGBoost model data is in good agreement

with the TCAD data (i.e.,  $I_{DS}$ - $V_{GS}$  and  $I_{DS}$ - $V_{DS}$ ) as shown in Fig.5(b-g).

#### 4. Mitigation of the DATA Imbalance

This section proposes a novel methodology for data augmentation using the utilization of latent representations derived from a stacked autoencoder (SAE), which serve as latent codes inside the InfoGAN framework (Fig.6a). The approach focuses on using the powerful representation capabilities offered by SAE to enhance the development of structured and semantical outputs.

The InfoGAN framework enhances the conventional generative adversarial network (GAN) architecture by incorporating a latent code, denoted as 'C', and thereafter aims to optimize the mutual information between this code and the generated data. (Fig.6b-e). Finally, the realized ML-based local simulator has been verified by calculating the timing parameter as tabulated in Table II.



**Fig. 6. (a) InfoGAN architecture: generator architecture: 256, 128, 64, 32 dense, respectively; discriminator architecture: 256 & 128 dense; SAE architecture: 128, 64, 32, 5 dense (latent codes), 32, 64, 128 dense, respectively. (b-d) distributions correspond to original data & distributions correspond to read data for drain current (b-c) and top radii of mid-sheet (d-e).**

Objective function of GAN:

$$\min_G \max_D V(G, D) = E_x \sim P_{data} [\log D(x)] + E_z \sim noise [\log (1 - D(G(z)))]$$

For a given generator the optimal discriminator is:

$$D(x) = \frac{P_{data}(x)}{P_{data}(x) + P_G(x)} \quad (3)$$

The objective function of InfoGAN where  $\lambda$  is a hyperparameter:

$$\min_{G,Q} \max_D V(D, G, Q) = V(G, D) - \lambda L_1(G, Q)$$

$$L_1(G, Q) \leq I(C; G(Z, C))$$

$$I(C; G(Z, C)) = H(C) - H(C|G(Z, C))$$

Here  $L_1(G, Q)$  stands for the lower bound of mutual information ( $I(C; G(Z, C))$ ).  $H$  stands for entropy.

## 5. Conclusion

This paper uses well-calibrated TCAD models to demonstrate the corner rounding effect in Nanosheet FET and provides a possible design guideline for optimum DC and RF performance. Further, the TCAD data is employed in training the machine learning model, i.e., the XGBoost regressor. The obtained RMS error is around  $5.0822 \times 10^{-7}$  concerning 500 testing samples and  $5.2994 \times 10^{-7}$  with unseen data. We further proposed the data augmentation technique using InfoGAN to suppress the data imbalance and enhance the model accuracy. Thus, Machine learning augmentation with TCAD assessment is worth exploring to achieve a time-efficient solution for device simulation, fault prediction, and optimization.

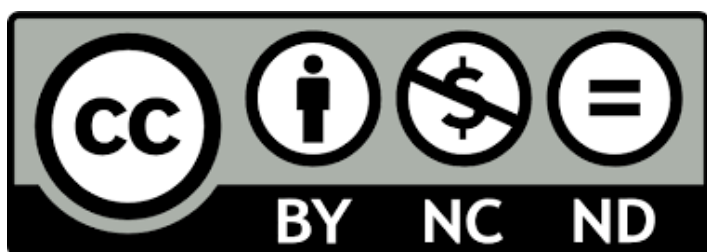
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